Sleep-HRV Interaction Analysis Using Data from Wearables

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ABSTRACT

This paper looks at the relationship between sleep metrics collected from wearable data and next day heart rate variability. The goal was to figure out which sleep variable carried the most information about next day heart rate variability. The main input variables were deep sleep duration, REM sleep duration and sleep efficiency. The target variable was next day average heart rate variability measured in milliseconds. The workflow covered data quality verification, daily aggregation, outlier analysis, exploratory data analysis, clustering, regression modelling, temporal validation and feature influence analysis. No missing values or duplicated rows were detected in the raw datasets or in the final model-ready dataset. Since heart rate variability is a continuous numerical variable the main modelling task was treated as regression. Several models were compared including baseline mean regression, linear regression, ridge regression, K-nearest neighbours regression, decision tree regression and random forest regression. The final selected model was K-nearest neighbours regression with k equal to 3 and distance-based weighting. It achieved MAE of 18.578, RMSE of 19.447 and R-squared of -62.734 on the final temporal test set. The negative R-squared value showed weak generalisation on unseen future data. For that reason the project was interpreted mainly as an exploratory wearable-data analysis rather than a reliable clinical prediction model. Under the combined ranking rule used in the notebook, deep sleep ranked first among the required sleep metrics, although the individual influence measures were not fully consistent and the result should be interpreted cautiously.

PROBLEM UNDERSTANDING AND DOMAIN BACKGROUND

Heart rate variability, also called HRV, measures the variation in time between consecutive heartbeats. It is often used as an indirect marker of autonomic nervous system regulation, recovery and physiological stress. Higher HRV is usually associated with stronger parasympathetic activity and better recovery while lower HRV may be related to stress, fatigue, illness or poor recovery [1]. HRV is also highly individual. For that reason changes over time within the same person are usually more meaningful than comparing one fixed HRV value between different people. Sleep is closely related to recovery. Deep sleep is usually connected with physical restoration, lower heart rate and increased parasympathetic activity. REM sleep is related to cognitive and emotional processing but it can also include more variable autonomic activity. Sleep efficiency describes how much of the time spent in bed was actually spent asleep. Because of these physiological relationships the project analysed whether deep sleep duration, REM sleep duration and sleep efficiency were associated with next-day HRV. Wearable devices allow this type of analysis because they collect sleep, heart rate and activity measurements in daily life. However wearable measurements should be interpreted carefully. Consumer wearables estimate sleep stages using indirect signals such as movement, photoplethysmography and device-specific algorithms. These estimates are not the same as clinical polysomnography or ECG-based HRV measurement [2]. For this reason this project was treated as exploratory and observational. The goal was not to prove that a sleep variable causes HRV changes. The goal was to analyse patterns in the available data and build a transparent modelling pipeline. The project also follows the main reporting logic of TRIPOD+AI for prediction models that use regression or machine learning methods [3]. Since feature importance was included the explainability part was reported with the main idea of CLIX-M in mind [4]. This means that the report clearly states what the explanation method measures and what it cannot prove.

1 Data and Preprocessing

Data Source

The project used wearable health data provided for the coursework project. Three CSV files were used: Vital Signs, Sleep and Activity. The available records covered the period from November 2025 to May 2026. The final dataset was created at the daily level, with one row representing one sleep-HRV observation.

The target variable was next-day average heart rate variability in milliseconds, stored in the notebook as `next_day_hrv_ms`. The primary predictors followed the project specification:

- `deep_sleep_min`: deep sleep duration in minutes
- `rem_sleep_min`: REM sleep duration in minutes
- `sleep_efficiency_pct`: sleep efficiency in percent

Additional variables, including total sleep time, awake time, light sleep, sleep-stage percentages, previous-day steps and previous-day calories, were kept for exploratory analysis. These variables were used to better understand the wearable data, but they did not replace the three required sleep predictors in the main regression model.

Cleaning and Daily Aggregation

The raw files were first checked for column names, data types, missing values and duplicated rows. Date columns were converted into datetime format and percentage variables were converted from text into numeric values, since fields like sleep ratio or oxygen saturation were originally stored as percentage strings. The sleep file contained individual sleep episodes with a start time, end time and wake-up time. Because the modelling target was daily HRV, each sleep episode was assigned to the date the person woke up. When more than one sleep record fell on the same wake-up date, the values were aggregated together. Sleep duration, awake time, REM sleep, light sleep and deep sleep were summed by wake-up date and sleep efficiency was recomputed after aggregation using:

$$\text{Sleep efficiency} = 100 \times \frac{\text{Time asleep}}{\text{Time in bed}}. \tag{1}$$

The vital signs data were then merged with the aggregated sleep data by date. Activity data were shifted by one day so that activity from day d-1 was assigned to the sleep and HRV observation on day d. This was done because activity from the previous day can influence recovery and next-day HRV.

Missing Values and Outliers

The raw Vital Signs, Sleep and Activity datasets were checked for missing values and duplicated rows. No missing values or duplicated rows were detected. The final model-ready dataset also contained no missing values. A `dropna()` operation was included in the notebook as a safety check for the required sleep variables and the HRV target, but the number of observations remained unchanged at 157 before and 157 after the check. Therefore, no observations were removed because of missing data and no missing-value imputation was performed. Outlier analysis was carried out using two methods. The interquartile range method flagged values below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$, while Z-score analysis flagged values with an absolute Z-score above 3. Outliers were not automatically removed since extreme HRV or sleep values can reflect real physiological variation rather than errors. Instead they were reported and used to get a sense of overall data quality.

Methods

Before moving on to modelling, exploratory data analysis was carried out to get a feel for the data. Descriptive statistics were used to look at the scale, spread and skewness of the variables. Time-series plots showed how sleep metrics and HRV changed over time, histograms were used to inspect distributions and scatter plots with trend lines were used to look at the relationship between each sleep metric and next-day HRV. Pearson correlation was used to rank the linear association between each variable and next-day HRV. This was not interpreted as causality, only as a first exploratory measure of association. A correlation heatmap was also produced to look at relationships between sleep variables, activity variables and HRV. Among the three required sleep inputs, REM sleep had the strongest absolute Pearson correlation with next-day HRV, though the association was still weak. The correlation values were:

- REM sleep: $r = -0.127$
- Deep sleep: $r = -0.065$
- Sleep efficiency: $r = -0.018$

All three correlations came out weak and negative in this dataset. This result was treated as a bivariate exploratory finding rather than the final word, since the actual feature influence conclusion was based on the combined interpretation of correlation, standardised coefficients, permutation importance and single-feature models. None of this means that more REM sleep or more deep sleep biologically lowers HRV, it only means that the observed linear association in this wearable dataset happened to be weakly negative. Beyond the three required sleep variables a wider feature correlation ranking was also computed, using extracted sleep-stage percentages, awake time, previous-day activity variables and vital sign variables. This wider ranking was used purely for interpretation and the final predictive model still relied on the three required sleep inputs since that directly followed the project task. What the wider ranking did help explain was why the required-only model had such limited predictive performance: next-day HRV likely depends on additional contextual factors that deep sleep, REM sleep and sleep efficiency alone do not fully capture.

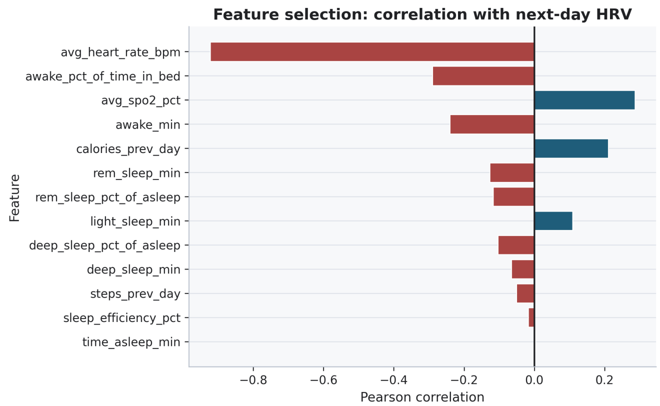

Figure 1. Feature correlation ranking for all candidate variables against next-day HRV.

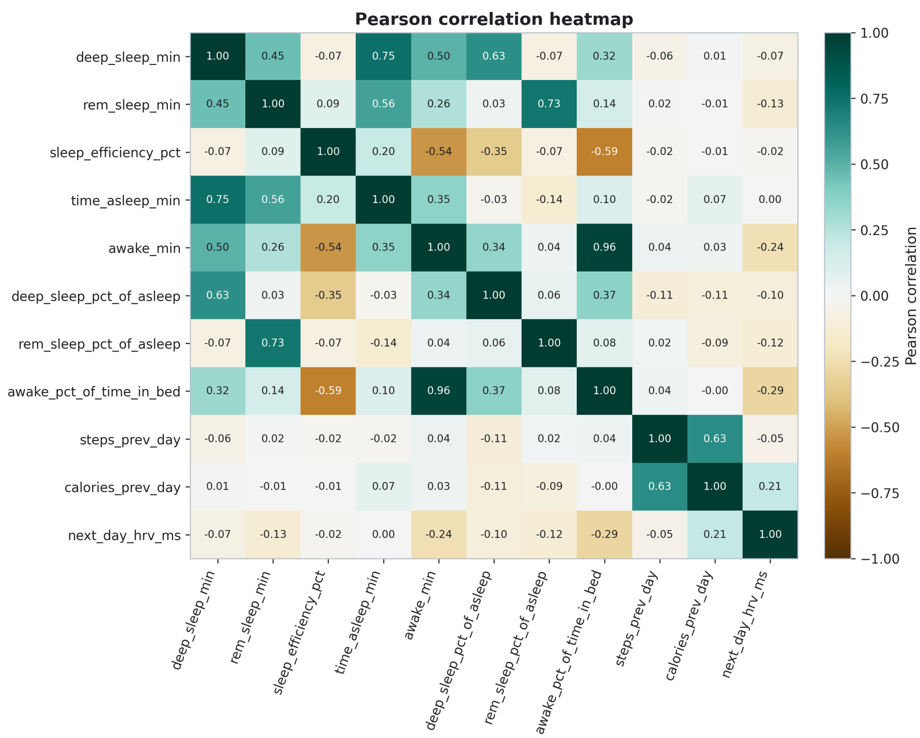

Figure 2. Pearson correlation heatmap for sleep, activity and HRV variables.

KMeans clustering is used as an exploratory method to identify sleep profiles based on deep sleep, REM sleep and sleep efficiency. The elbow method and silhouette score were used to inspect the possible number of clusters. The clustering was not used as the final prediction model. Its purpose was to just check whether nights with similar sleep structure also had different average next day HRV.

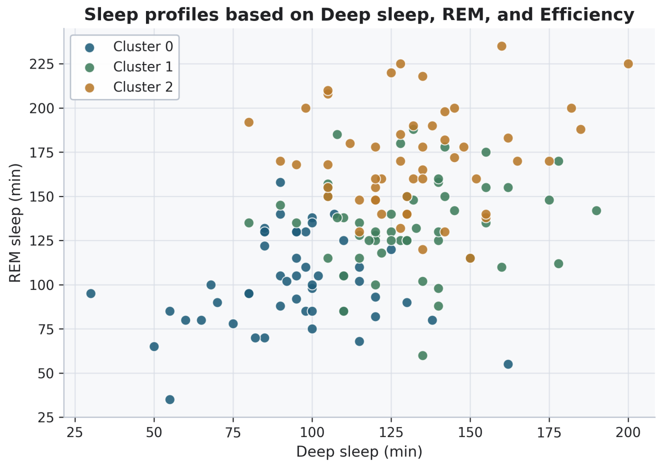

Figure 3. KMeans sleep-profile clusters created from deep sleep, REM sleep and sleep efficiency, visualised using deep sleep and REM sleep axes.

Modelling Approach

Since next-day HRV is a continuous numerical target, the main task here was regression. Each daily observation was treated as a pattern represented by the feature vector:

$$x = [\text{deep sleep}, \text{REM sleep}, \text{sleep efficiency}]. \quad (2)$$

Several regression models were compared, each included for a specific reason. The baseline mean model simply predicted the average training HRV for every observation, giving a reference point for whether the other models were actually learning anything useful. Linear regression was used as a simple statistical baseline, learning coefficients for the sleep features and producing an interpretable prediction of continuous next-day HRV. Ridge regression added regularisation on top of that which made sense given the dataset was small and some predictors were likely correlated. K-nearest neighbours regression was included as a distance-based nonparametric method, since it predicts HRV from the most similar previous nights and ties directly into pattern recognition ideas like distance, similarity and nearest-neighbour learning. Decision tree regression brought in an inductive rule-based approach that could capture simple nonlinear relationships between sleep variables and HRV. Random forest regression was included as an ensemble method to see whether combining multiple trees improved stability compared with a single tree. The model that ended up being selected was K-nearest neighbours regression with k=3 and distance-based weighting. For a new observation this model looks at the three most similar previous nights in the training data and predicts HRV as a distance-weighted average of their HRV values, so nights that are closer in feature space carry more influence on the prediction

Validation

Because the data are daily wearable observations ordered in time, a temporal train, validation and test split was used rather than a random one. Random splitting was avoided since it could let future observations leak into training and make the evaluation unrealistic. The first 60 percent of observations went into training, the next 20 percent into validation and the final 20 percent into testing. The final modelling dataset contained 157 daily observations. Because no missing values were present in the required predictors or the target, all 157 observations were retained. The temporal split contained 94 observations for training, 31 for validation and 32 for testing. Models were first fitted on the training set and compared on the validation set and the model with the lowest validation RMSE was selected. After that the best model was refitted on the combined train and validation data and evaluated once on the unseen temporal test set. As an additional robustness check, 5-fold time-series cross-validation was also run, where the training window was gradually expanded over time and each model was tested on a later fold. This helped check whether model performance held up across different temporal splits rather than relying on just one fixed division. The cross-validation comparison covered linear regression, ridge regression, K-nearest neighbours regression and decision tree regression. The following regression metrics were used:

- MAE: mean absolute error
- RMSE: root mean squared error

- R^2: coefficient of determination

MAE gives the average absolute prediction error in milliseconds, RMSE penalises large errors more heavily and R^2 shows whether the model explains more variation than a simple mean baseline would. A negative R^2 means the model actually performs worse than just predicting the mean target value.

Results

Model Selection Results

Based on validation RMSE, K-nearest neighbours regression with $k = 3$ came out as the best model. This selected model was then evaluated on the temporal test set.

Table 1. Final Test Performance of the Selected Model

Metric	Value
MAE	18.578 ms
RMSE	19.447 ms
R^2	-62.734

Although KNN with $k = 3$ had the best validation RMSE, the final test R^2 turned out to be strongly negative which means the model did not generalise well to future unseen data. The modelling pipeline itself was methodologically sound, but the selected model was not reliable for accurate HRV prediction in this dataset.

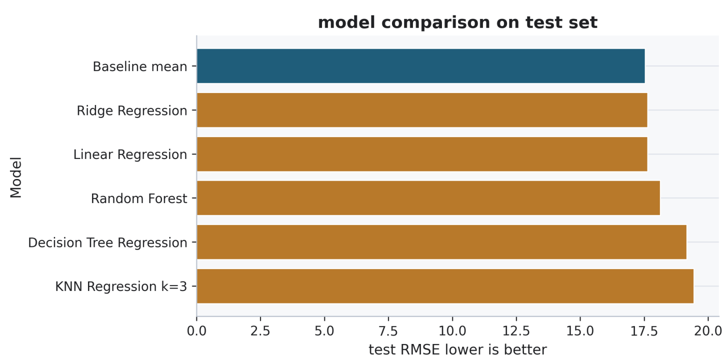

Figure 4. Comparison of regression models on the test set using RMSE.

Residual analysis showed that prediction errors varied across the test period. The model made large errors on some days. This supported the conclusion that the prediction model was not stable enough for reliable use.

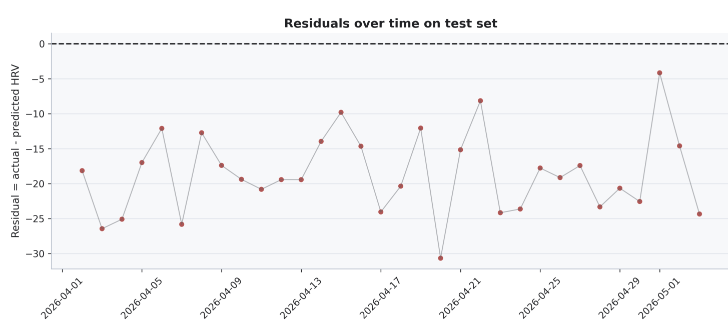

Figure 5. Residuals of the selected model over the temporal test set.

Feature Influence Results

Feature influence was assessed using Pearson correlation, standardised linear regression coefficients, permutation importance and single-feature models. These were combined because each one captures a different part of the picture: correlation measures direct linear association, standardised coefficients measure scaled linear effects, permutation importance shows how much the selected model actually depended on each variable and single-feature models test how much predictive value each required sleep metric carries on its own. Taken together, this analysis pointed to deep sleep duration as the most influential required sleep metric, meaning it carried the most useful information for next-day HRV among the three required inputs in this dataset. That said, since the final model had weak test performance, this should be read as an exploratory finding rather than a settled result. It points to a pattern worth discussing, not a clinically reliable rule.

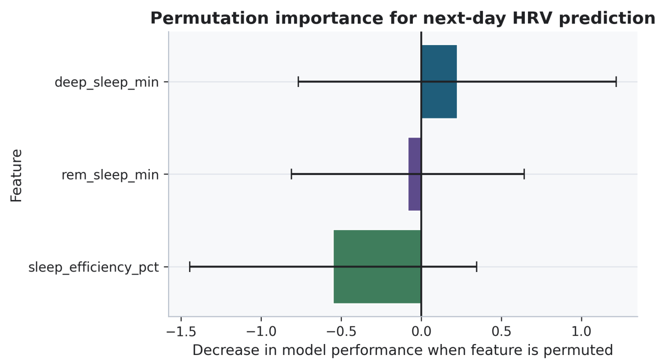

Figure 6. Permutation importance for the selected HRV prediction model.

Discussion

The main goal of this project was to figure out which sleep metrics have the strongest influence on next-day HRV. What the results showed is that this relationship is not simple, at least not in this wearable dataset. Pearson correlations between the required sleep metrics and HRV came out weak and the regression models also generalised poorly on the temporal test set. Taken together this suggests that deep sleep, REM sleep and sleep efficiency on their own were not enough to accurately predict next-day HRV. The final selected model was K-nearest neighbours regression with k=3. From a pattern recognition standpoint this made sense since it relies on similarity between daily sleep patterns. That said KNN tends to be sensitive to small datasets, local noise and shifts in distribution over time which may explain why it performed best on the validation set but fell apart on the final test set. The negative R^2 means that the model performed worse than a mean-value baseline on unseen future data. That is still a meaningful finding though, since it shows the available sleep variables had limited predictive value for future HRV in this particular dataset. In a health-related prediction task poor generalisation should be reported plainly rather than glossed over. The feature influence analysis pointed to deep sleep duration as the most informative of the required sleep metrics. This makes physiological sense given that deep sleep is often tied to physical recovery and parasympathetic activity. Even so it should be read with some caution, since deep sleep can also be tangled up with things the dataset never captured, like training load, illness, stress, caffeine, alcohol, menstrual cycle, bedtime regularity or sleep timing. The clustering results were useful for describing sleep profiles but they do not amount to causal evidence. KMeans can surface groups of similar nights but those groups depend heavily on which variables were selected and how they were scaled. So any HRV differences between clusters are best treated as descriptive patterns rather than anything stronger.

Limitations

A few limitations are worth mentioning. First the dataset was small, with only 157 daily observations in the final modelling set which is on the thin side for machine learning. Second the data came from a wearable device rather than clinical ECG and polysomnography and while wearables are great for everyday tracking their sleep-stage estimates can carry real measurement error.

Third the analysis was observational in nature. It can point to associations and predictive patterns but it cannot prove that any one sleep metric actually causes changes in HRV. Fourth the main model relied on only three required sleep predictors and HRV is shaped by plenty of factors the dataset simply did not capture, things like stress, illness, exercise intensity, hydration, alcohol, caffeine, medication, menstrual cycle and sleep regularity.

Finally the explainability results speak to model behaviour rather than physiology. Permutation importance shows how much the trained model leans on a given feature, but that should not be read as direct biological causation.

Conclusion

This project developed an exploratory wearable data pipeline for analysing the relationship between sleep metrics and next day HRV. The workflow included data quality verification, daily aggregation, outlier analysis, exploratory data analysis, sleep-profile clustering, regression modelling, temporal validation and feature influence analysis. No missing values were detected, so no observations were removed or imputed because of missing data.

The final selected model was K-nearest neighbours regression with $k = 3$ and distance-based weighting. It was selected using validation RMSE. On the final unseen test set, the model achieved MAE of 18.578 ms, RMSE of 19.447 ms and R^2 of -62,734. The negative R^2 showed poor generalisation, so the model should not be interpreted as a reliable clinical prediction model.

The most useful result was the feature influence analysis. The feature-influence methods produced different rankings. REM sleep showed the strongest absolute Pearson correlation and the best single-feature test performance, sleep efficiency had the largest absolute standardised linear coefficient and deep sleep was the only feature with positive mean permutation importance. Under the combined ranking rule used in the notebook, deep sleep was ranked first. However, this result should be interpreted as an exploratory model-based finding rather than evidence of a causal relationship.

References

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